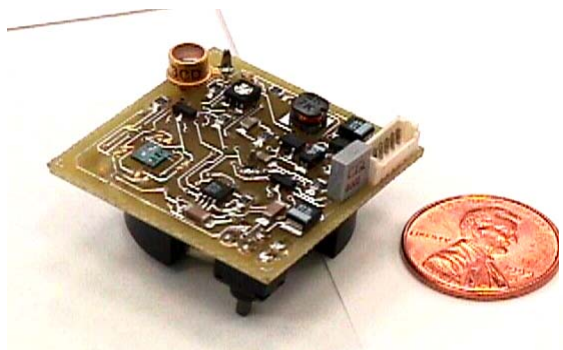
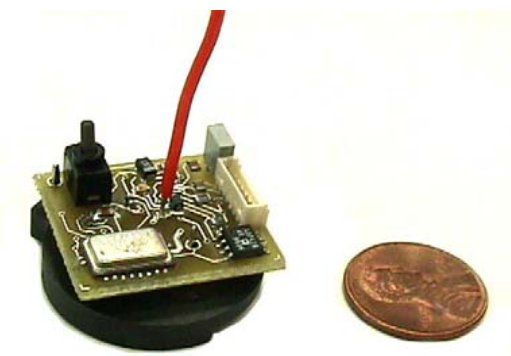
Sensor Network

Technologies and Algorithms





Introducing Sensor Network 1/2

- Definition of sensor networks
  - Large number of small and low cost sensor nodes
    - sensing, processing, and wireless communication capabilities
  - Densely deployed inside/close to the phenomenon
  - Node position not engineered or predetermined
    - Deployment in inaccessible terrain or disaster relief
    - Protocols and algorithms with self-organization capabilities
    - Nodes have to cooperate and partially process sensed data

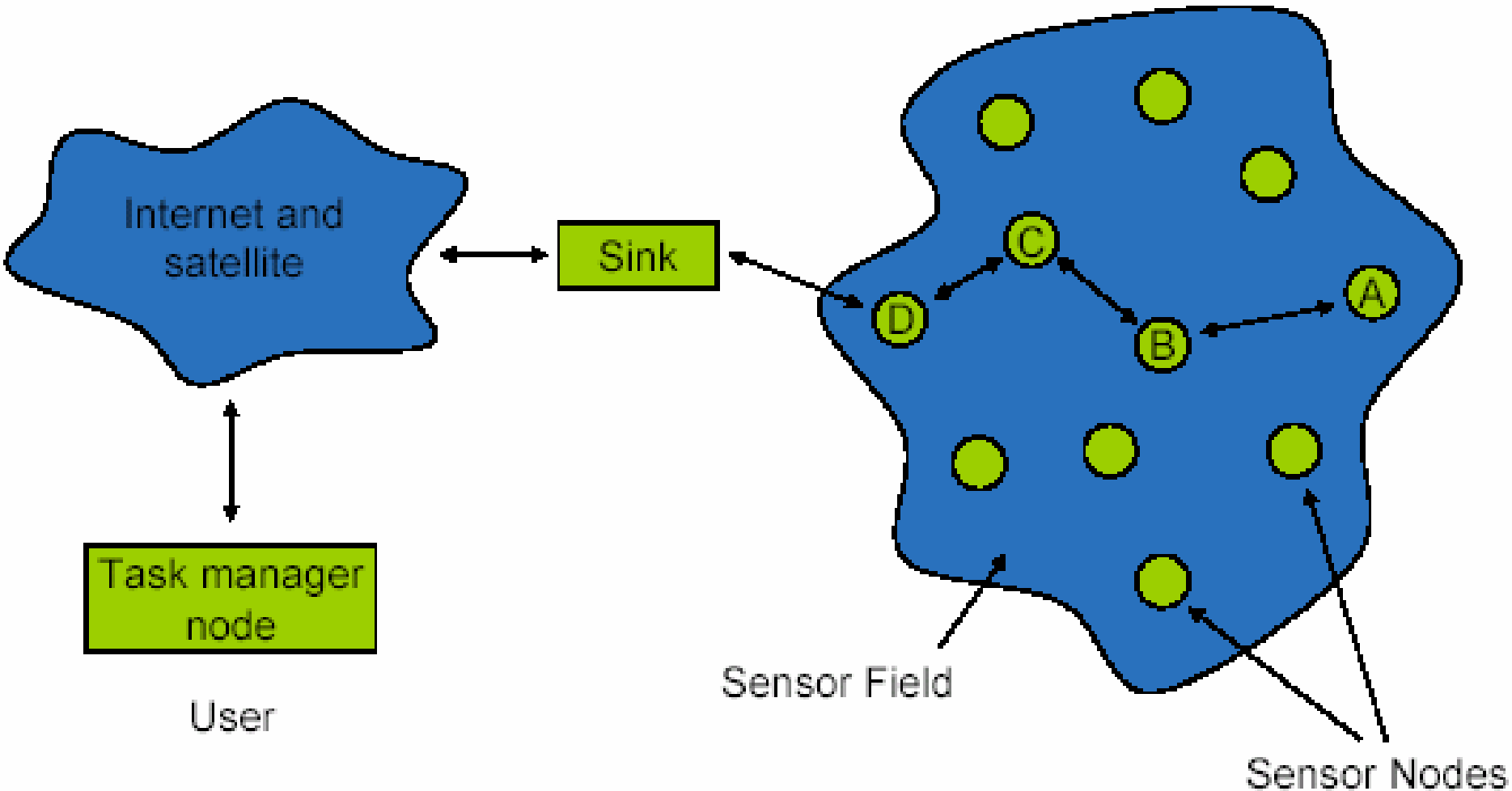




Introducing Sensor Networks 2/2

- Sensor Types
  - Seismic
  - Magnetic
  - Thermal
  - Visual
  - Infrared
  - Acoustic
  - Radar
- Sensing
  - Continuous
  - Event detection
  - Location sensing
  - Actuator control
- Monitored ambient conditions
  - Temperature
  - Humidity
  - Vehicular movement
  - Lightning condition
  - Pressure
  - Soil makeup
  - Noise levels
  - Presence/absence of objects
  - Mechanical stress level
  - Object speed, direction, size

Typical Sensor Network Architecture



Sensor Network Applications

- Environmental applications
  - Biology, meteorology, geophysics
  - Agriculture
  - Forest fire detection
  - Flood detection
- Health applications
  - Interfaces for the disabled
  - Tele-monitoring of human physiological data
  - Drug administration in hospitals
- Home applications
  - Home automation
  - Smart environment
- Other commercial applications
  - Environmental control in office buildings
  - Interactive museums
  - Monitoring car thefts
  - Managing inventory control
- Military Applications

Examples

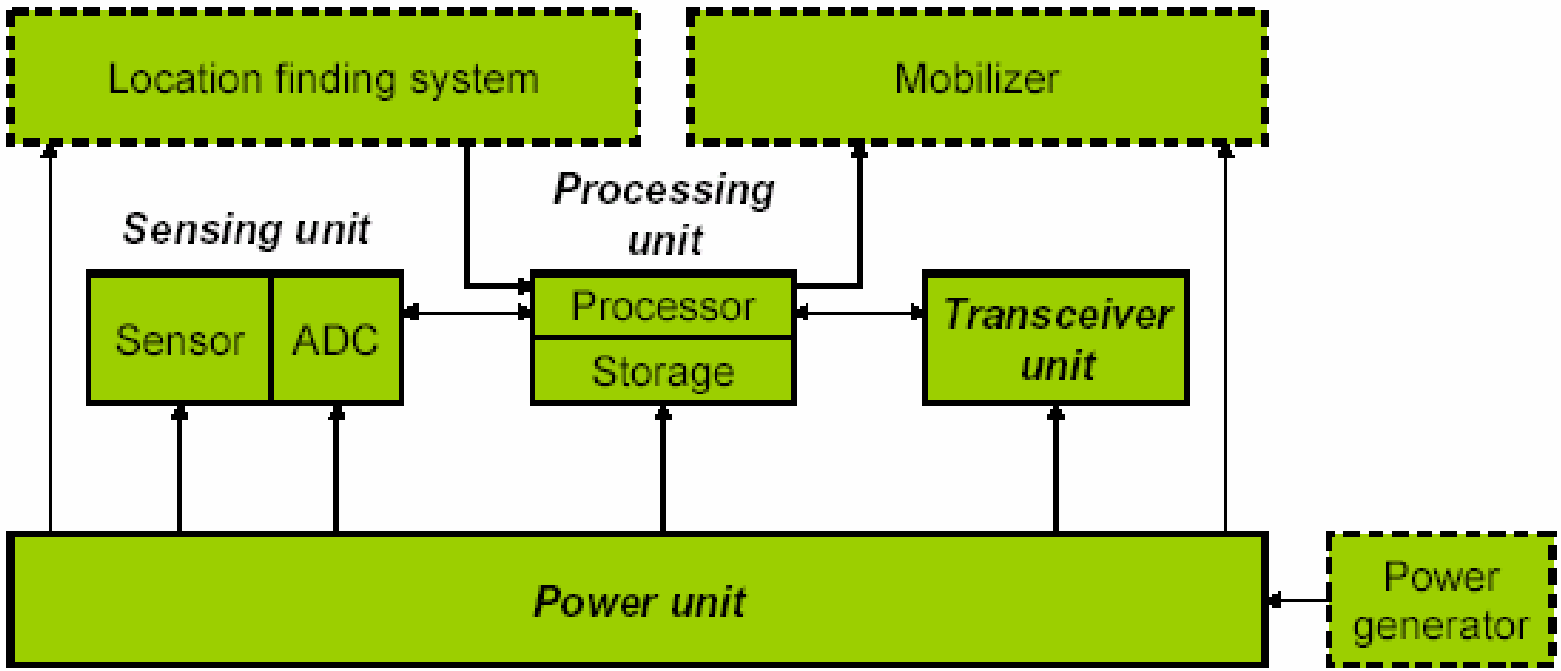


- Monitoring Birds activity at Great Duck Island (CA, USA).
  - Dilemma for Biologists
    - Need multiple measurements of biological parameters at frequent intervals
    - potentially harming their subjects and biasing results
  - Solution: "Mote Sensing", using small wireless probes
    - Array of individual Motes, capable of recording temperature, humidity, pressure, and other environmental data
    - Allows to follow nesting activity throughout the season with minimal impact on the birds
- Smart Buildings
  - Make buildings, bridges, and other structures aware of their own health
  - Traditional solutions end up in jungle of wires
- Avalanche Tracking

Sensor Network VS. MANET

- Special class of ad-hoc network
- Most ad-hoc network techniques not well suited
- Difference to ad-hoc networks
  - Number of nodes several orders of magnitude higher
  - Sensor nodes are deployed densely
  - Sensor nodes are prone to failures
  - Frequent topology changes
  - Communication mainly based on broadcast paradigm
  - Limited power, computational capabilities, and memory
  - Sensor nodes have no global identification
  - Energy-power is critical

Components of a Sensor Node



Design Factors 1/4

- Fault tolerance
  - Sensor node may fail or be blocked
    - Lack of power
    - physical damage
    - environmental interference
  - Failure of individual node should not affect the complete network
- Scalability
  - Number of nodes may reach an extreme value of millions
  - Node density may be in order of hundreds in a region
  - Schemes must be scalable and utilize high node density

Design Factors 2/4

- Production Cost
  - Cost of single node very important to justify cost of the network
    - Otherwise traditional tethered sensors would be the alternative
  - Sensor node should be less than 1€
    - E.g. Bluetooth 10 times more expensive than the targeted price
- Hardware constraints
  - Sensor node subunits need to fit in a matchbox-sized module
    - Required size may be smaller than a cubic centimeter
    - Light enough to remain suspended in the air
  - Additional constraints
    - Extreme energy efficient
    - Low production cost, dispensable
    - Autonomous, operate unattended, adaptive to the environment

Design Factors 3/4

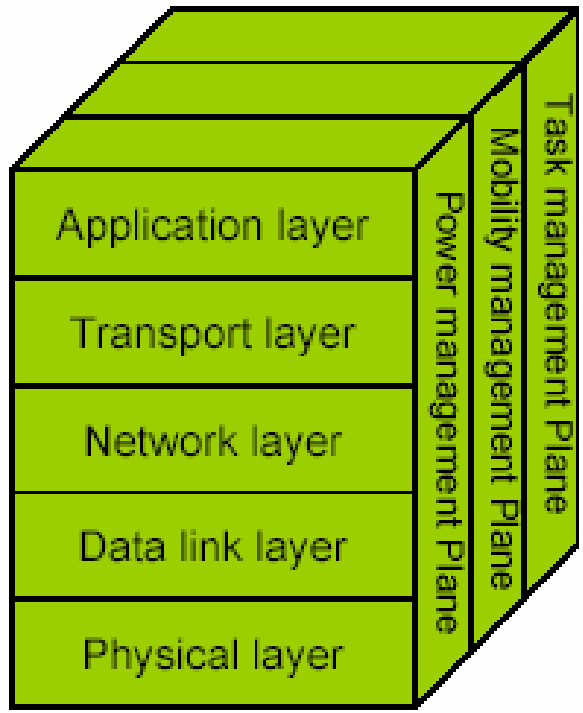
- Sensor network topology
  - Pre-deployment and deployment
    - Thrown in as a mass or placed one by one
  - Post-deployment
    - Topology changes due to position changes, reachability, available energy, malfunctioning, task details
  - Redeployment of additional nodes
    - Redeployment at any time to replace malfunctioning nodes or due to changes in task dynamics
- Environment
  - Home or large building, interior of a large machinery, bottom of an ocean, contaminated field,

Design Factors 4/4

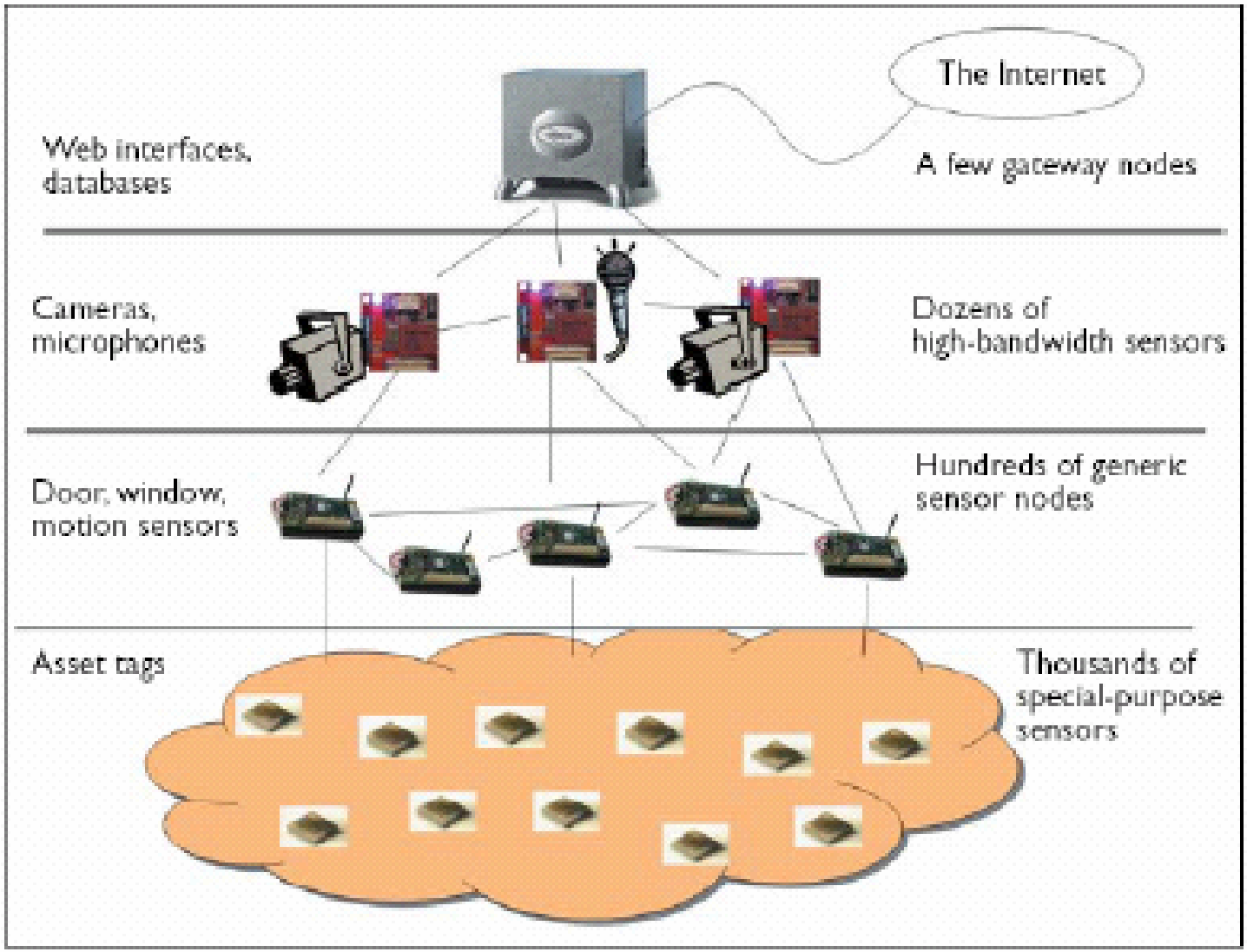
- Transmission media
  - Radio
    - Used by much of the current hardware
    - Must be available worldwide (e.g. 2.4GHz, or 916MHz)
  - Infrared, or optical media:
    - License-free, robust to interference from electrical devices, cheaper
    - Line of sight between sender and receiver
- Power consumption
  - Limited power sources: <0.5Ah, 1.2V
  - Replenishment of power source might be impossible
  - Sensor node lifetime coupled with battery lifetime
  - Power consumption in three domains: sensing, communication, and data processing

Protocol Stack

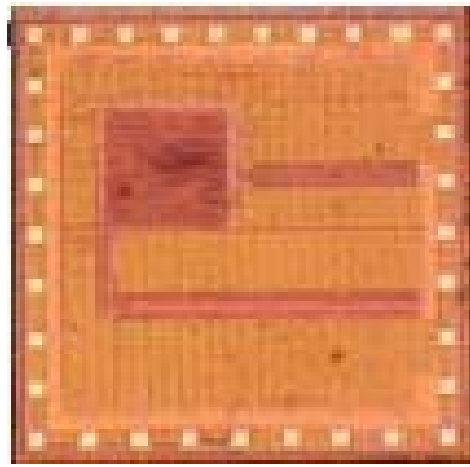
- Management planes
  - Coordinate sensing task and lower overall power consumption
- Power management
  - E.g. turn off receiver after message receipt
  - Disconnect from routing task due to low power
- Mobility management
  - Track movement of nodes in order to maintain routes back to the user
- Task Management
  - Schedule sensing task to a specific region
  - Nodes with more power are used more frequent



Sensor Node Hierarchy



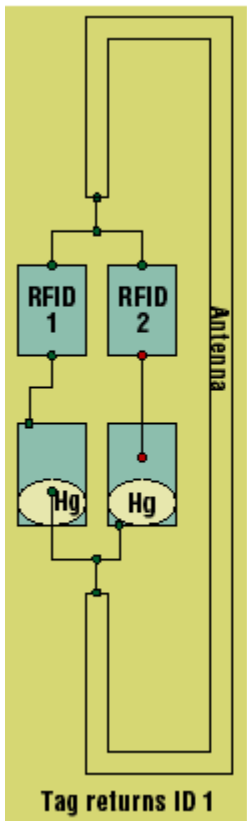
Special Purpose Sensor Node 1/2

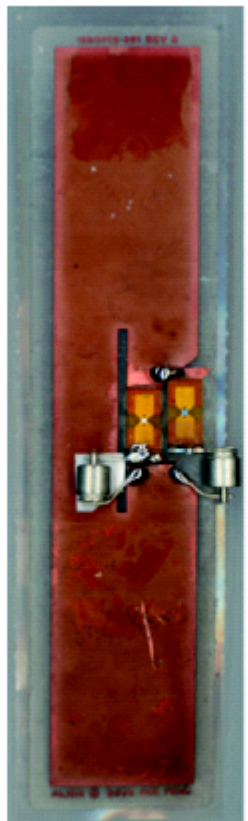


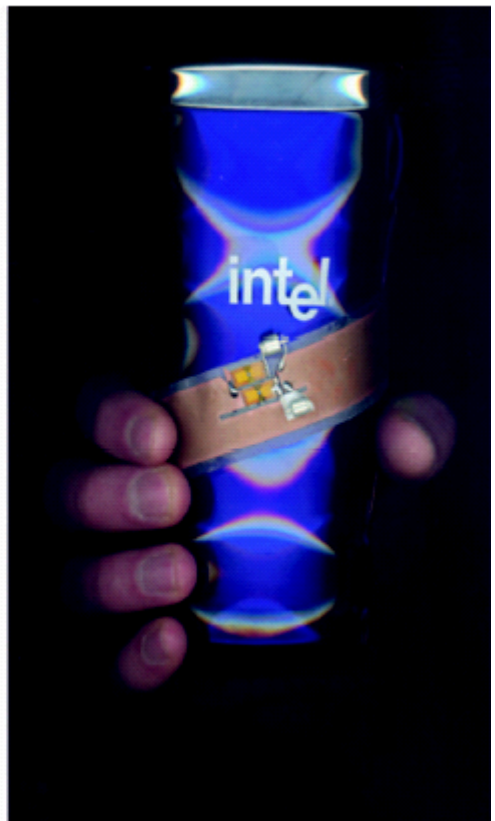
- Cubic-millimeter-scale devices (see Smart Dust)
- Extremely limited energy resources
- Typical duty cycle 0.1%-0.5%
- Example scenario: track mobile assets
  - Trigger an alarm when asset leaves facility without authorization
  - Periodically report its presence for years
- Example: Spec node
  - Single-chip node for ultra low cost and low power consumption
  - 2.5 mm x 2.5 mm
  - Includes data RAM (< 4Kb), minimal onboard processing, and communication
  - Can interface only with simple sensors; Specialized low bandwidth sampling or advanced RF tag
  - Communicate over short distance (Bandwidth <50 kbps)
  - Current version has only transmitter (future work: transceiver)

Special Purpose Sensor Node 2/2

- Example: Sensor based RFID
  - Passive technology
  - Require centralized readers (Proxy Idea)

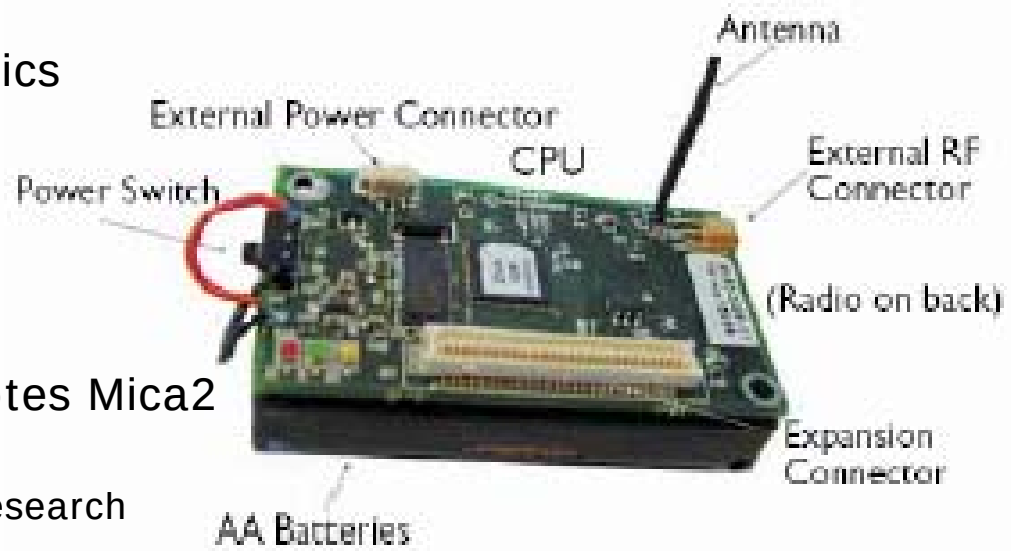






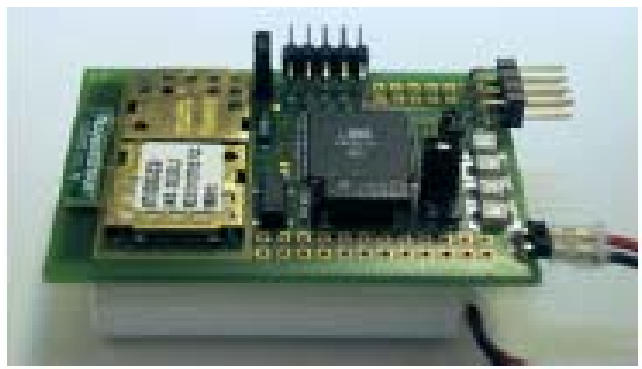
Generic Sensor Node

- Simple and specific function
- Require long term battery operation
- Typical duty cycle 1%-2%
- E.g. sensors placed on windows and doors for intrusion detection
- Typical operating characteristics
  - Size 1-10cm^3
  - General-purpose sensing and communications relay
  - Bandwidth <100kbps
  - Flash <0.5Mb ,RAM <10kb
- Notable example: Berkley motes Mica2
  - Off-the-shelf components
  - Most popular sensor network research platform
  - Can be connected with a wide range of sensors



High-Bandwidth Sensor Nodes

- Handle high bandwidth of data coming from complex sensors (video, acoustic, vibration, …)
- May require battery power but often plugged into public power system for long-term operation
- Example iMote (Intel)
  - Bluetooth transceiver (~500Kbps)
  - On chip RAM ~128Kb
- BlueDrake (Embit)



Gateway Sensor Nodes

- End-point for mesh of sensor nodes
  - Containing database/aggregation software to process and store individual sensor readings
- Provide an interface into many existing network types
- Example platform Star-gate (Intel)
  - 400 MHz X-scale architecture
  - Megabytes of RAM
  - Gigabytes of persistent storage
  - Capable to interface directly to Mica2 and iMote
  - Bridging the data to 802.11, Ethernet, …
  - Can provide a Web front-end to the sensor network

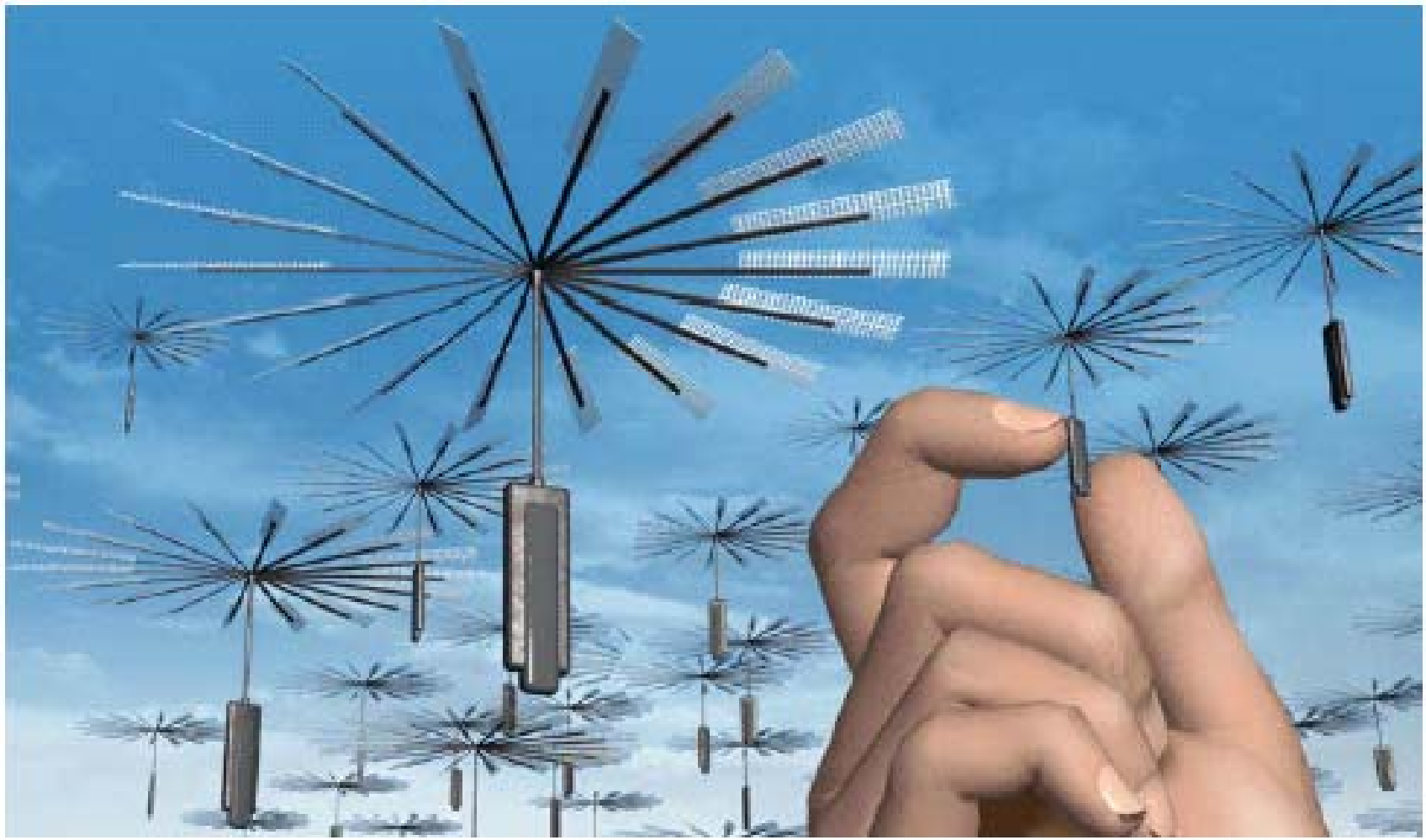
Current Sensor Network Platform 1/2

Node	CPU	Power	Memory	I/O and Sensors	Radio	Remarks
Special-purpose Sensor Nodes						
Spec 2003	4–8Mhz Custom 8-bit	3mW peak 3uW idle	3K RAM	I/O Pads on chip. ADC	50–100Kbps	Full custom silicon. traded RF range and accuracy for low-power operation.
Generic Sensor Nodes						
Rene 1999	ATMEL 8535	.036mW sleep 60mW active	512B RAM 8K Flash	Large expansion connector	10Kbps	Primary TinyOS development platform.
Mica-2 2001	ATMEGA 128	.036mW sleep 60mW active	4K RAM 128K Flash	Large expansion connector	76Kbps	Primary TinyOS development platform.
Telos 2004	Motorola HC50B	.001mW sleep 32mW active	4K RAM	USB and Ethernet	250Kbps	Supports IEEE 802.15.4 standard. Allows higher-layer Zigbee standard. 1.8V operation
Mica-Z 2004	ATMEGA 128		4K RAM 128K Flash	Large expansion connector	250Kbps	Supports IEEE 802.15.4 standard. Allows higher-layer Zigbee standard.

Current Sensor Network Platform 2/2

Node	CPU	Power	Memory	I/O and Sensors	Radio	Remarks
High-bandwidth Sensor Nodes						
BT Node 2001	ATMEL Mega 128L 7.328Mhz	50MW idle 285MW active	128KB Flash 4KB EEPROM 4KB SRAM	8-channel 10-bit A/D, 2 UARTS Expandable connectors	Bluetooth	Easy connectivity with cell phones. Supports TinyOS. Multihop using multiple radios/nodes.
Imote 1.0 2003	ARM 7TDMI 12-48MHz	1mW idle 120mW active	64KB SRAM 512KB Flash	UART, USB, GPIO, I^2C, SPI	Bluetooth 1.1	Multihop using scatternets, easy connections to PDAs, phones, TinyOS 1.0, 1.1.
Gateway Nodes						
Stargate 2003	Intel PXA255		64KNSRM	2 PCMICA/CF, com ports, Ethernet, USB	Serial connection to sensor network	Flexible I/O and small form factor power management.
Inrysnc Cericube 2003	Intel PXA255		32KB Flash 64KB SRAM	Single CF card, general-purpose I/O		Small form factor, robust industrial support. Linux and Windows CE support.
PC104 nodes	X86 processor		32KB Flash 64KB SRAM	PCI Bus		Embedded Linux or Windows support.

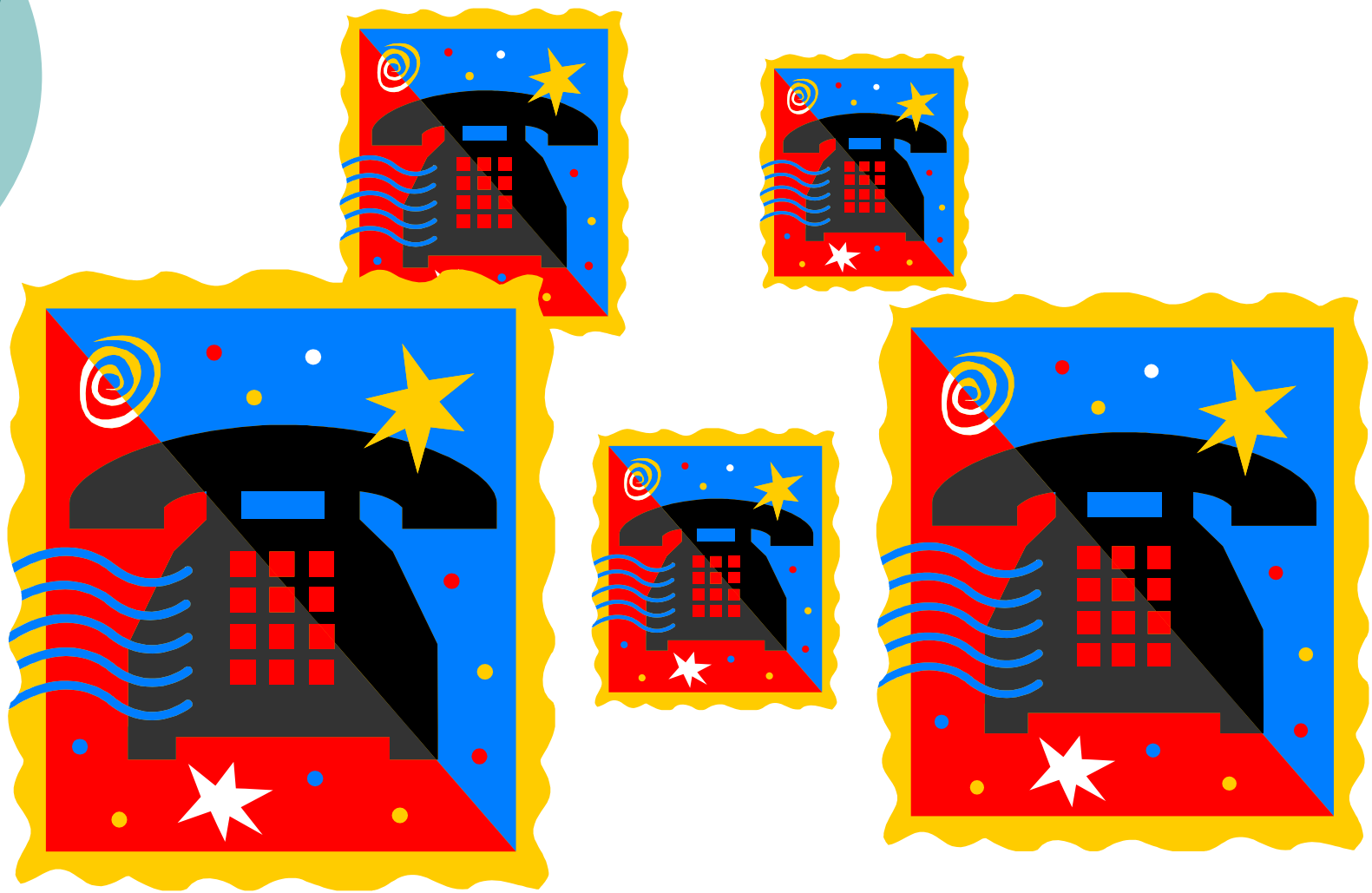
A Vision Of the Future…



Sensor Network Key Algorithms

- Communication
  - Sensor nodes have no unique address. How to communicate?
- Localization
  - In a number of application the physical location of nodes is fundamental. How can nodes discover such information?
- Clock Synchronization
  - Sensor clock synchronization is fundamental to synchronize duty cycle and to perform collective inference. How can be achieved?
- Global Programming
  - How can the sensor network be programmed efficiently?

Data-Driven Communication

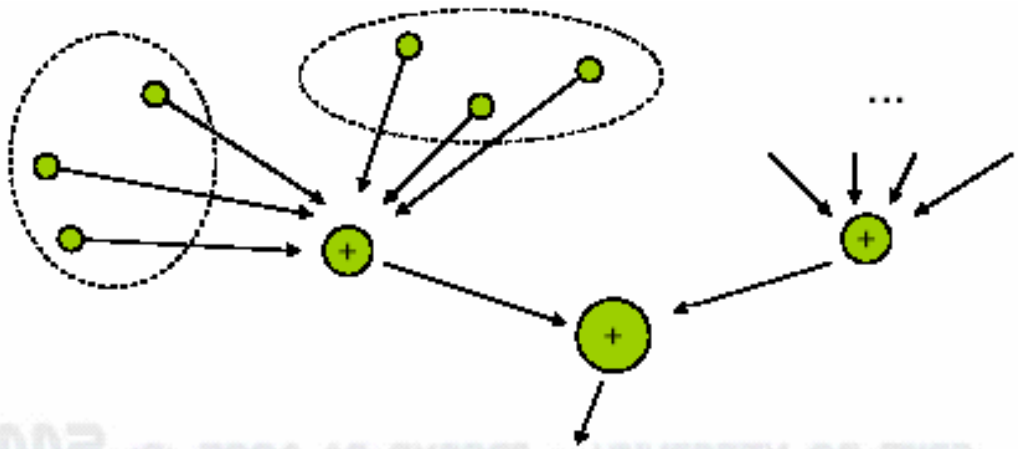


Data-Driven Communication in Sensor Network

- Classic Communication Patterns
  - Unicast, Broadcast, Multicast
  - Addressing of individual node or set of individual nodes
- What if individual nodes disappear?
- Data-driven communication
  - Tuple-space
  - P2P Overlay Network
  - Events
- Two possible Data-Centric Routing Approaches
  - Disseminate interest about data
  - Advertise available data and wait for request
- Can be position-based and topology-based or both

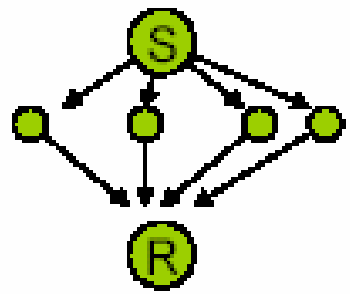
How to Address Nodes with no ID?

- Attribute-Based naming
  - Rather query an attribute than an individual node
  - E.g. the areas where the temperature is over 50°C, all available information about a running application in a certain area, …
- Data aggregation often needed to merge data received from many nodes (data fusion)
  - Some specifics may not be left out (e.g. location of the data)

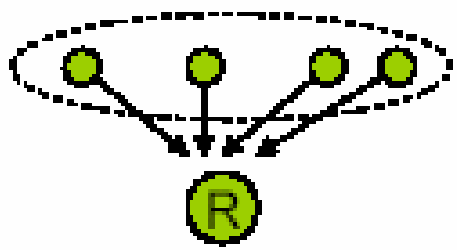


Flooding and Gossiping

- Flooding: When receiving packet for the first time, repeat forwarding, if maximum hop or destination not reached
  - Reactive technique
  - Does not require costly topology maintenance
- Deficiencies
  - Implosion
  - Overlap
  - Resource Blindness: Does not take energy resources into account
- Gossiping: forward to one random selected neighbor only
  - Avoids implosion problem
  - Message propagation takes a long time



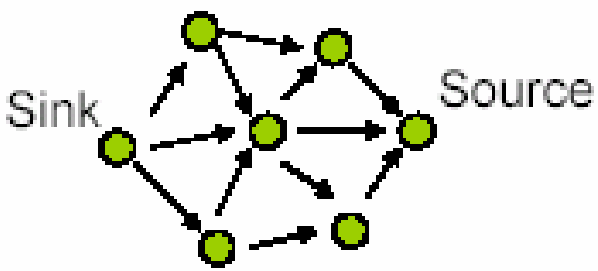
Implosion Problem



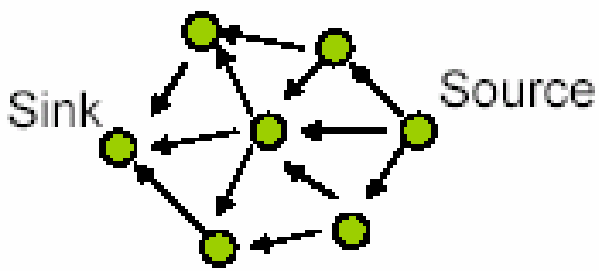
Overlap Problem

Directed Diffusion

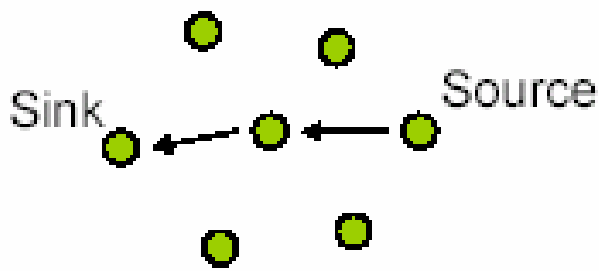
- Set up gradients for data to flow from source nodes to interested sink node
- Sink sends out interest
  - Attribute value pairs describing sensing task
- Interest Propagated through the network
  - Cached in each node to build gradients back to the sink
- Data from sources is sent back along interest gradient paths
  - Data aggregation performed locally at intermediate nodes
- Paths that provide valuable information and reinforced



(a) Propagate interest



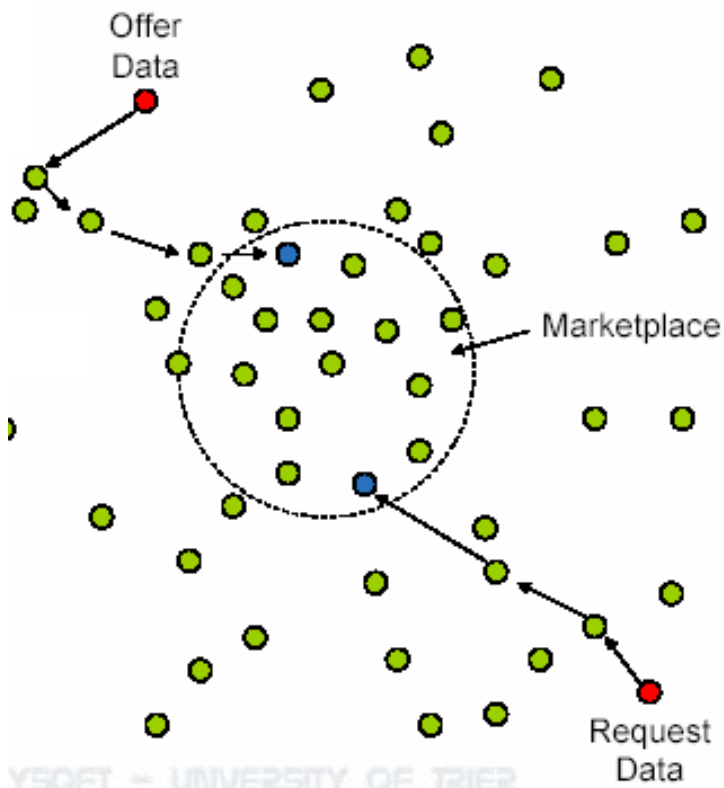
(b) Set up gradient



(c) Send data

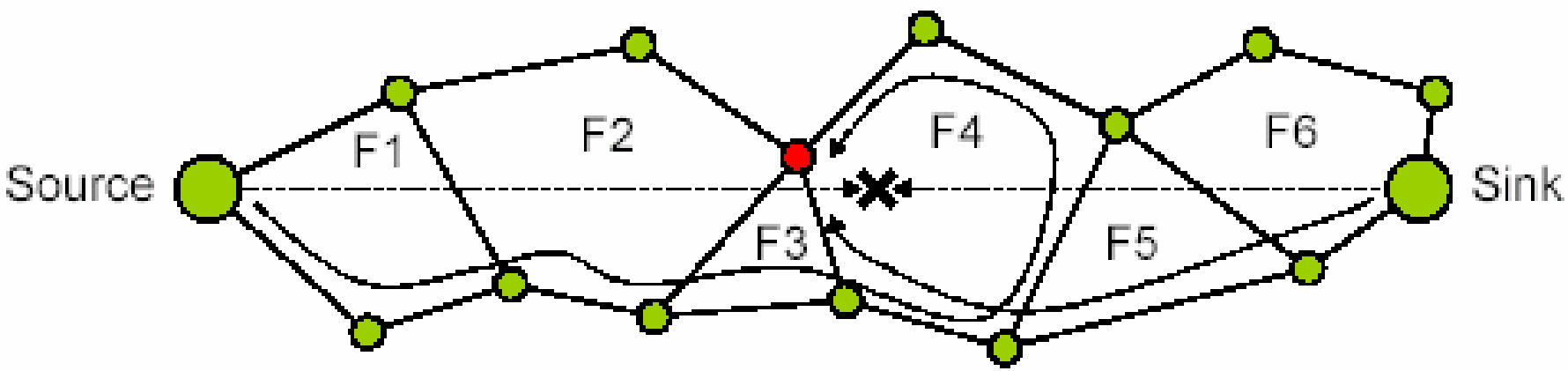
Geocasting

- Reach nodes in a certain area
- Geocasting Components
  - Routing towards the area
    - Single-path, multi-path
    - Restricted directional flooding
  - Dissemination inside the area
    - Location-aware flooding
    - Reducing redundant transmissions
- Marketplace Pattern
  - Place offer in a geographic area with high node density
  - Send request towards the same area

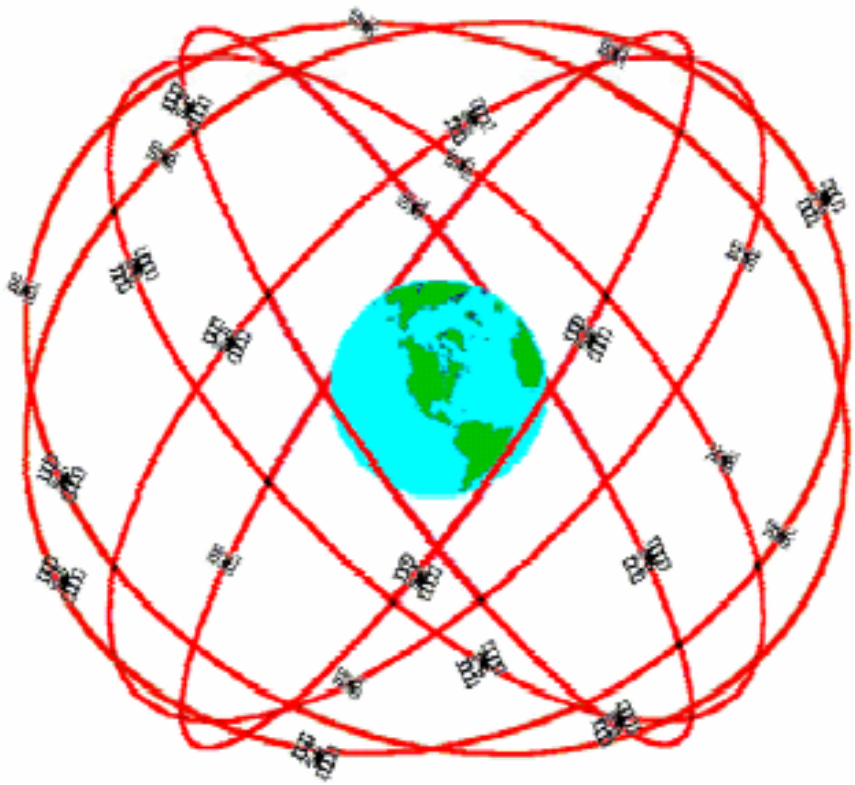


GHT: Geographic Hash Table

- Idea: hashing on geographical positions
- Put() and Get() operations map to the same device near to the hashed location
- Mapped device stores data
- Use of planar graph routing to find the same device



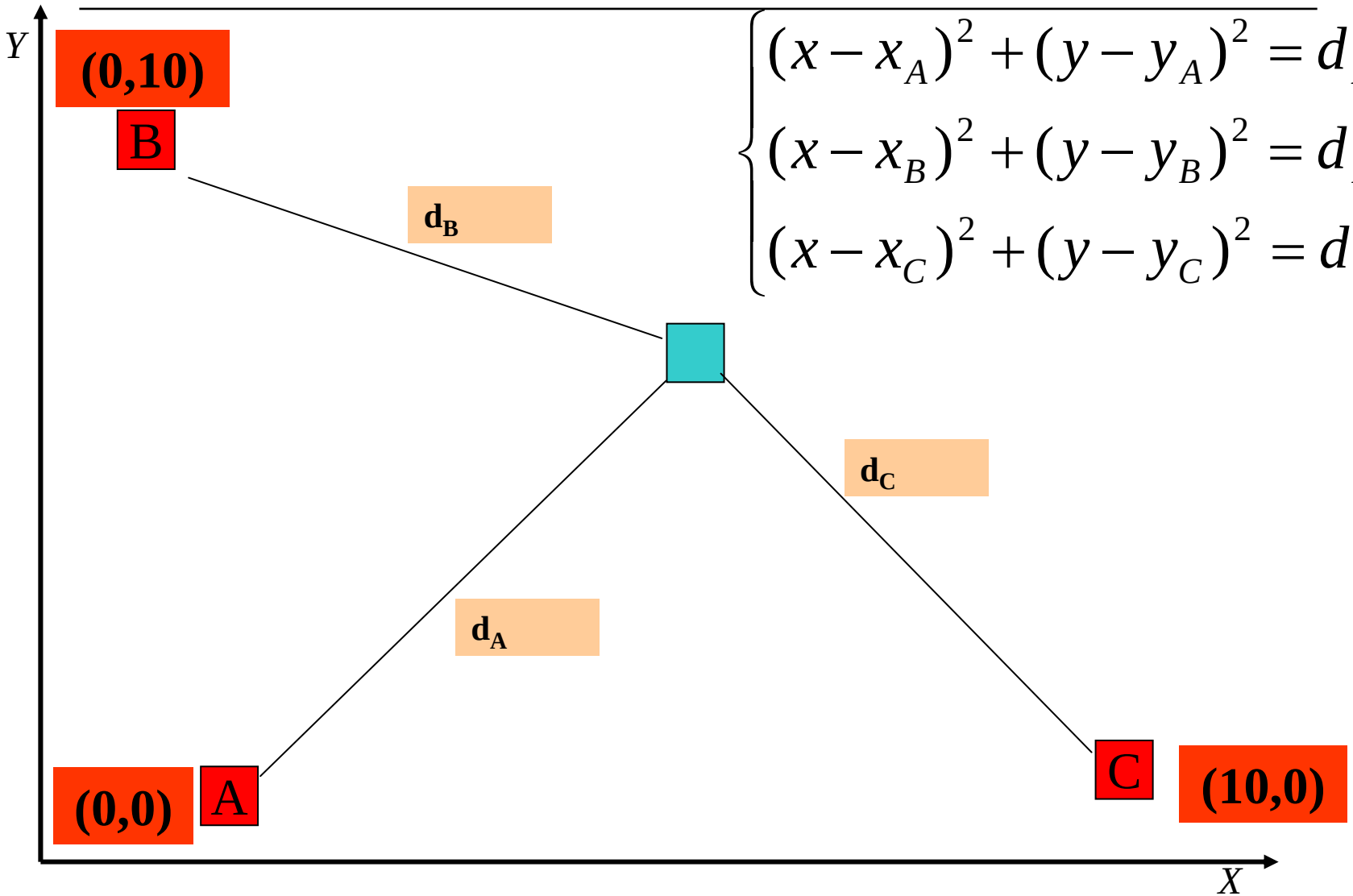
Spatial Localization In Sensor Network



Localization

- Most Important Contextual Information in a number of application
  - Spatial Information
  - Geographic Routing
- There are two main techniques to localize
  - Triangulation
    - Measure distance from multiple reference points and triangulate
  - Scene Analysis
    - Scene observed from particular point
    - Objects presence sensed using a physical phenomenon with limited range

Triangulation



$$\begin{cases} (x-x_A)^2+(y-y_A)^2=d_A^2 \\ (x-x_B)^2+(y-y_B)^2=d_B^2 \\ (x-x_C)^2+(y-y_C)^2=d_C^2 \end{cases}$$

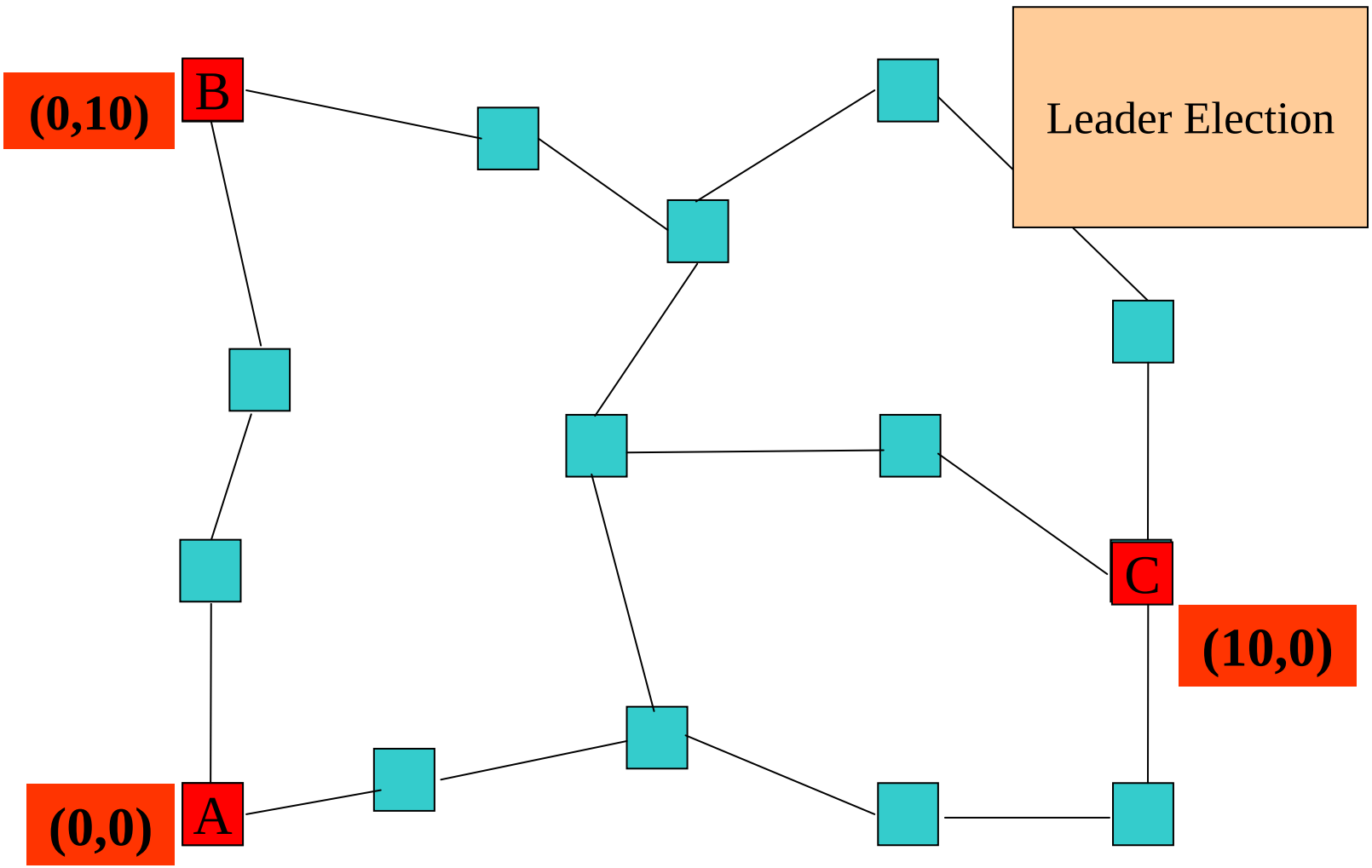
Localization via triangulation in Sensor Network

- Principle
  - Beacons at known coordinates emit a signal to neighbor devices (e.g., those in the communication range)
  - Devices determines distance on the basis of specific properties of this signal
- Mechanisms
  - Signal attenuation (e.g., radio signal)
    - Not enough precision on short distances
  - Travel time of acoustic waves or phase displacement of IR waves
    - Could be more accurate (cm-scale obtained with acoustic waves and IPAQs)
    - Requires very accurate time synchronization (which is another challenging problem by its own)
  - **Density based approaches**

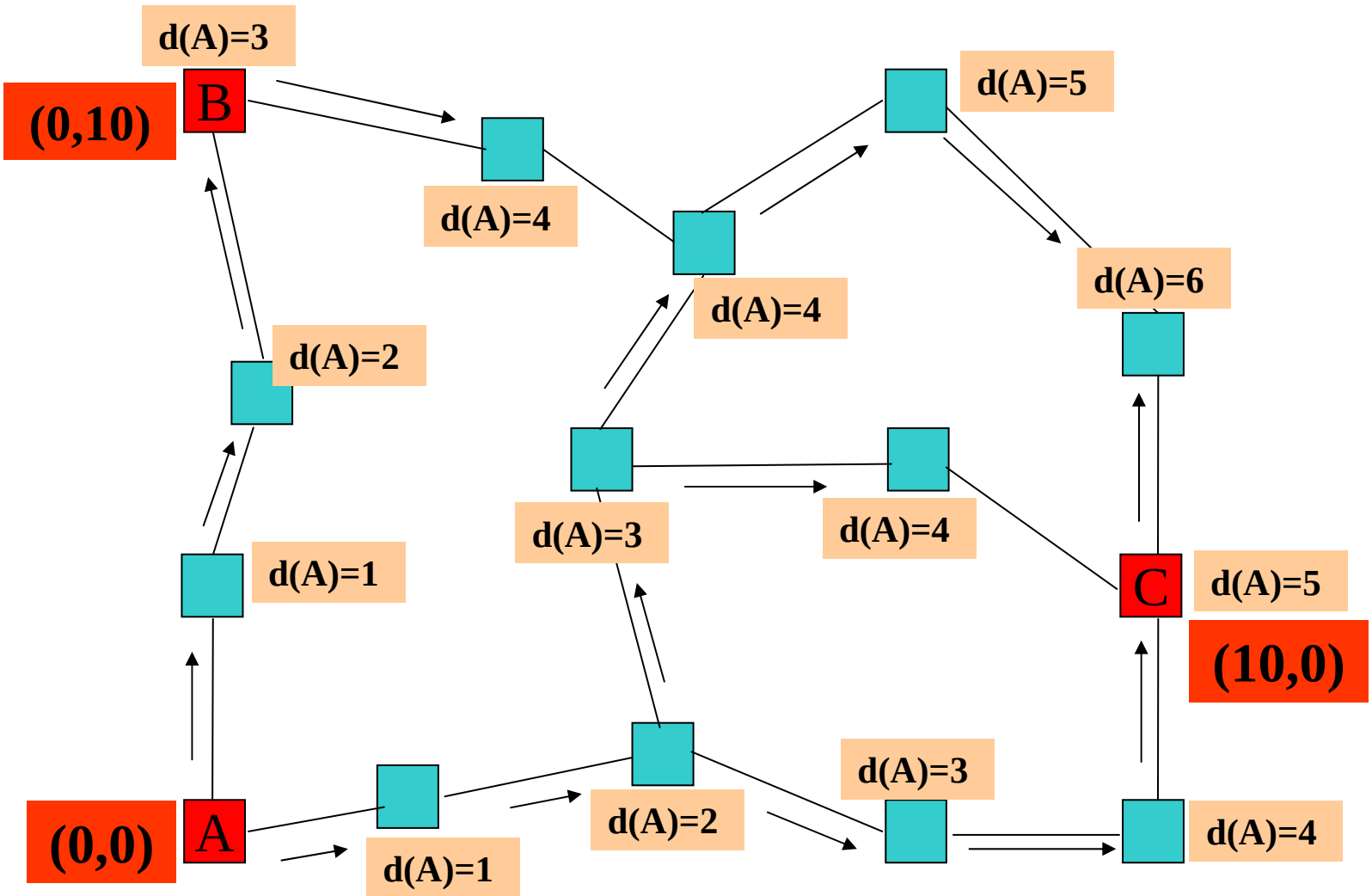
Density-based Distance Estimation

- Requires Devices
  - to be “dense” and to know their average density
  - to be able to send short-range signals to close devices
- Then
  - The beacon send a message to neighbors with a integer = 1, counting the number of hops the message has traveled
  - The signal is re-transmitted by each device after having incremented the counter
  - After a while, devices know their distance, in terms of hops, from the beacons
- Eventually
  - Triangulate and Calculate *Hops*AvgDist = PhysDist*
- Accuracy
  - Depends on average density: *Accuracy=AvgDist*
  - Requires at least 15 one-hop neighbors
  - Could be improved by multiple iterations

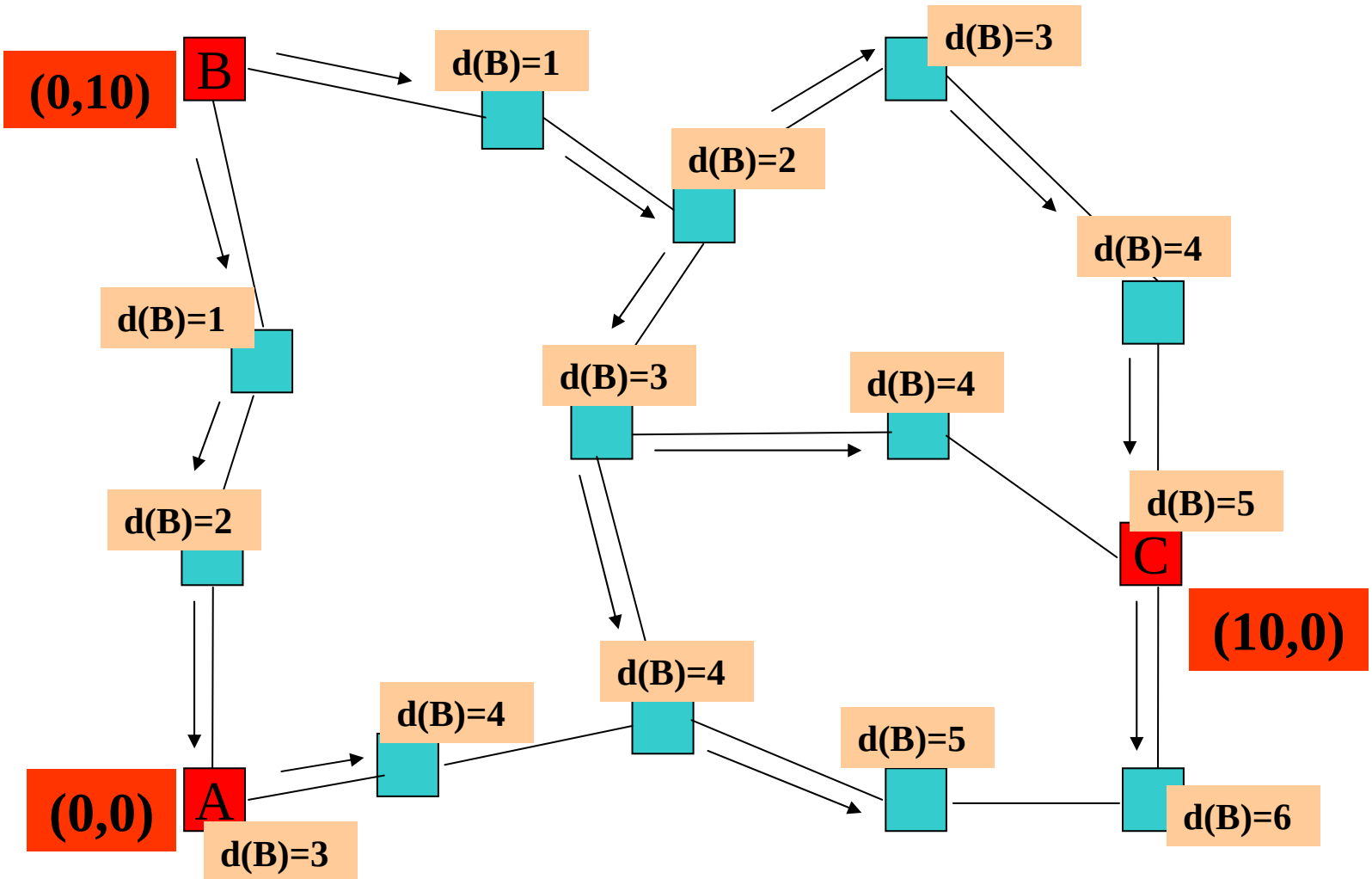
Example



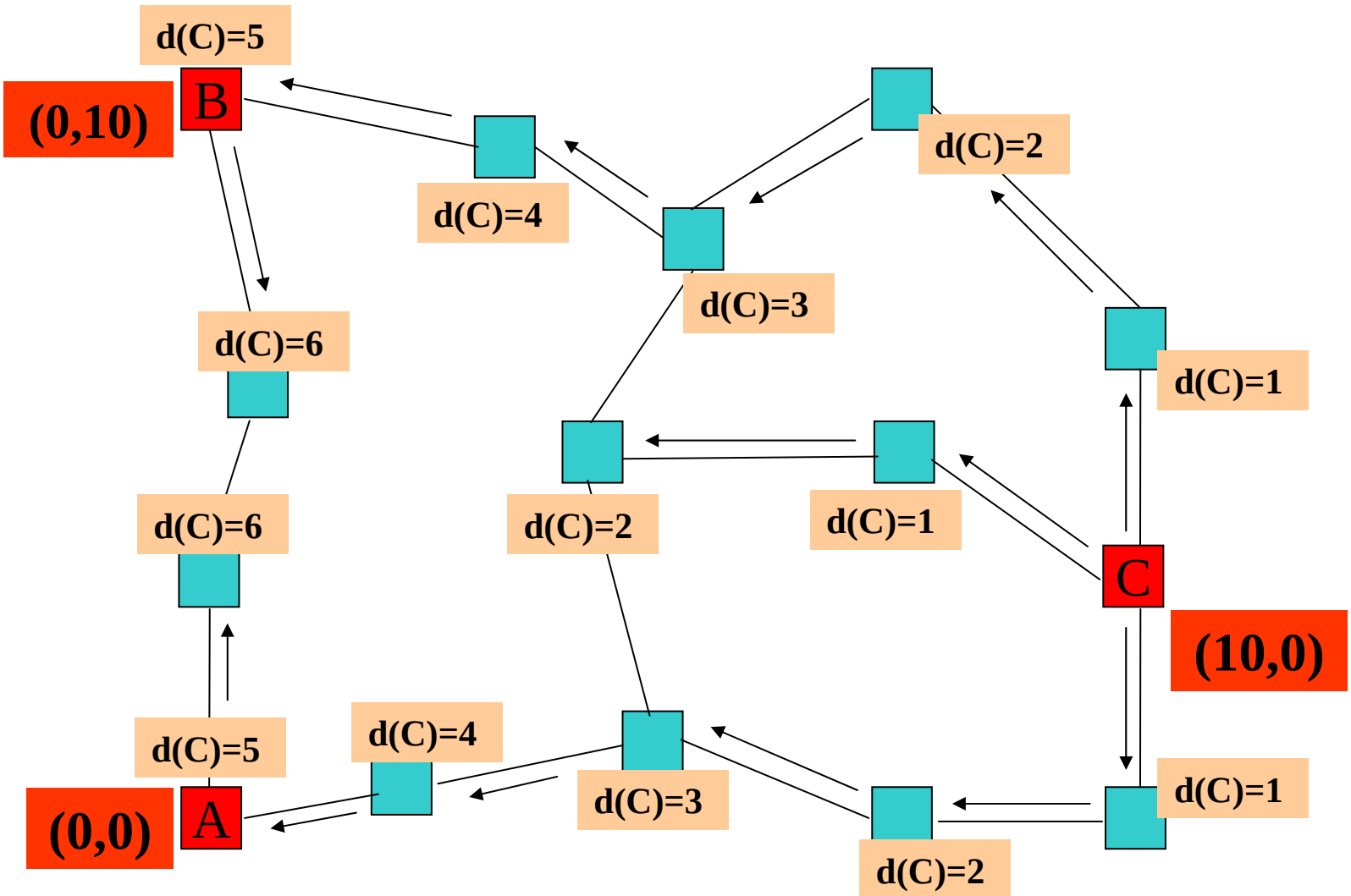
Example


d(A)=3
(0,10)
B
d(A)=4
d(A)=5
d(A)=4
d(A)=6
d(A)=2
d(A)=3
d(A)=4
d(A)=1
C
d(A)=5
(10,0)
(0,0)
A
d(A)=1
d(A)=2
d(A)=3
d(A)=4

Example

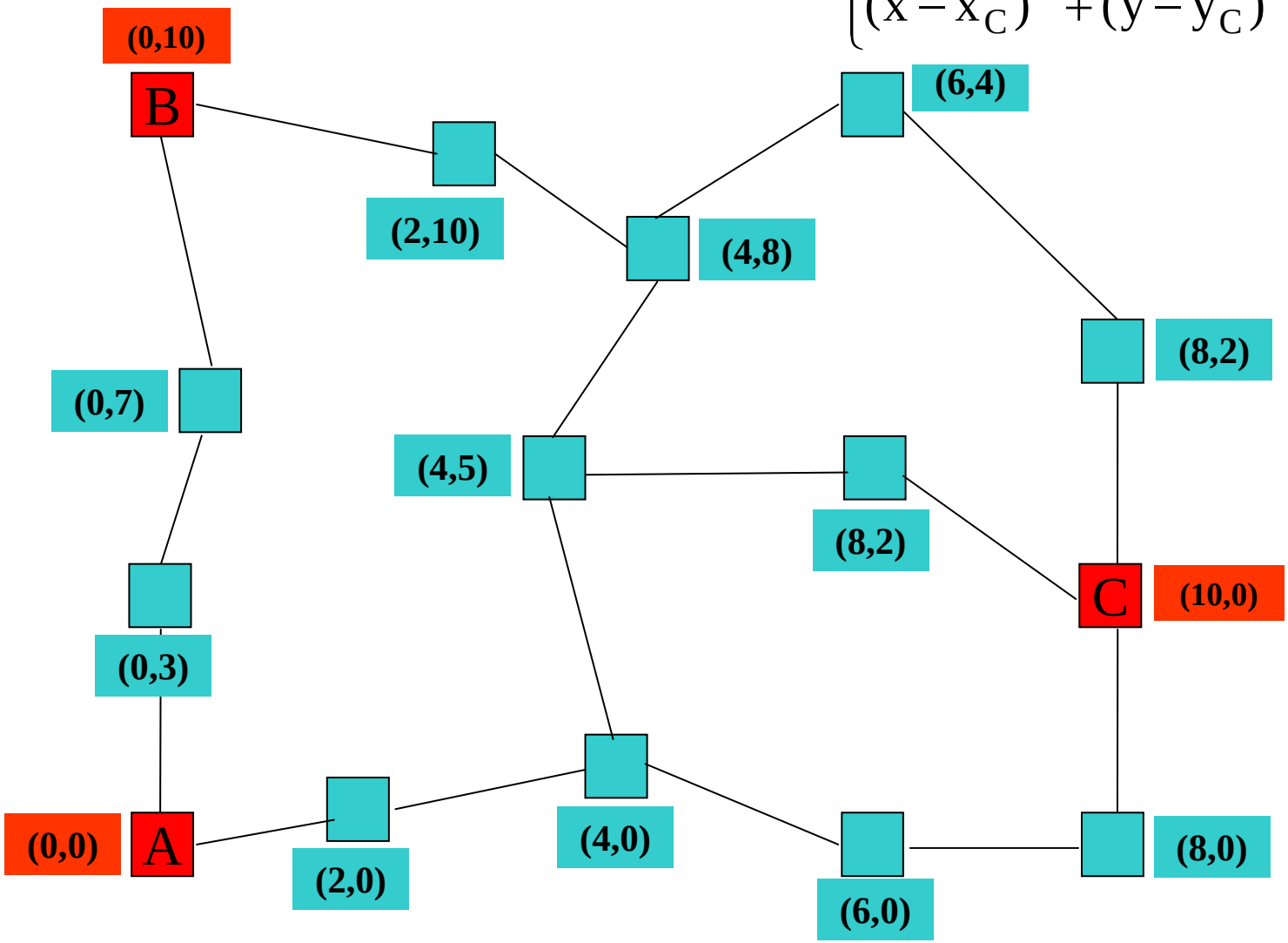

(0,10)
B
d(B)=1
d(B)=3
d(B)=2
d(B)=4
d(B)=1
d(B)=3
d(B)=4
d(B)=2
d(B)=5
C
(10,0)
d(B)=4
d(B)=4
d(B)=5
(0,0)
A
d(B)=6
d(B)=3

Example

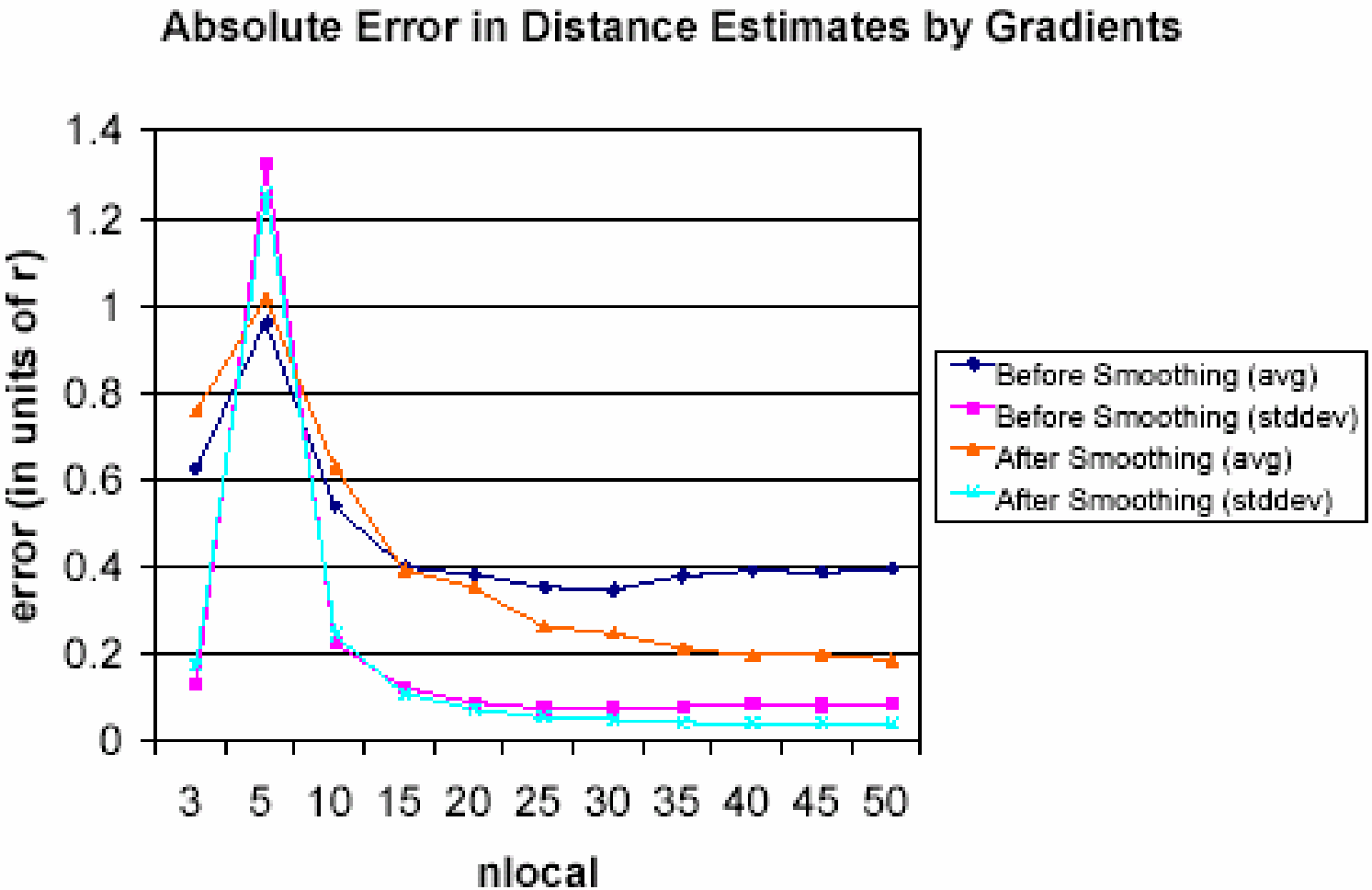

d(C)=5
(0,10)
B
d(C)=4
d(C)=2
d(C)=3
d(C)=6
d(C)=1
d(C)=6
d(C)=2
d(C)=1
C
(10,0)
d(C)=5
d(C)=4
d(C)=3
d(C)=1
(0,0)
A
d(C)=2

Example

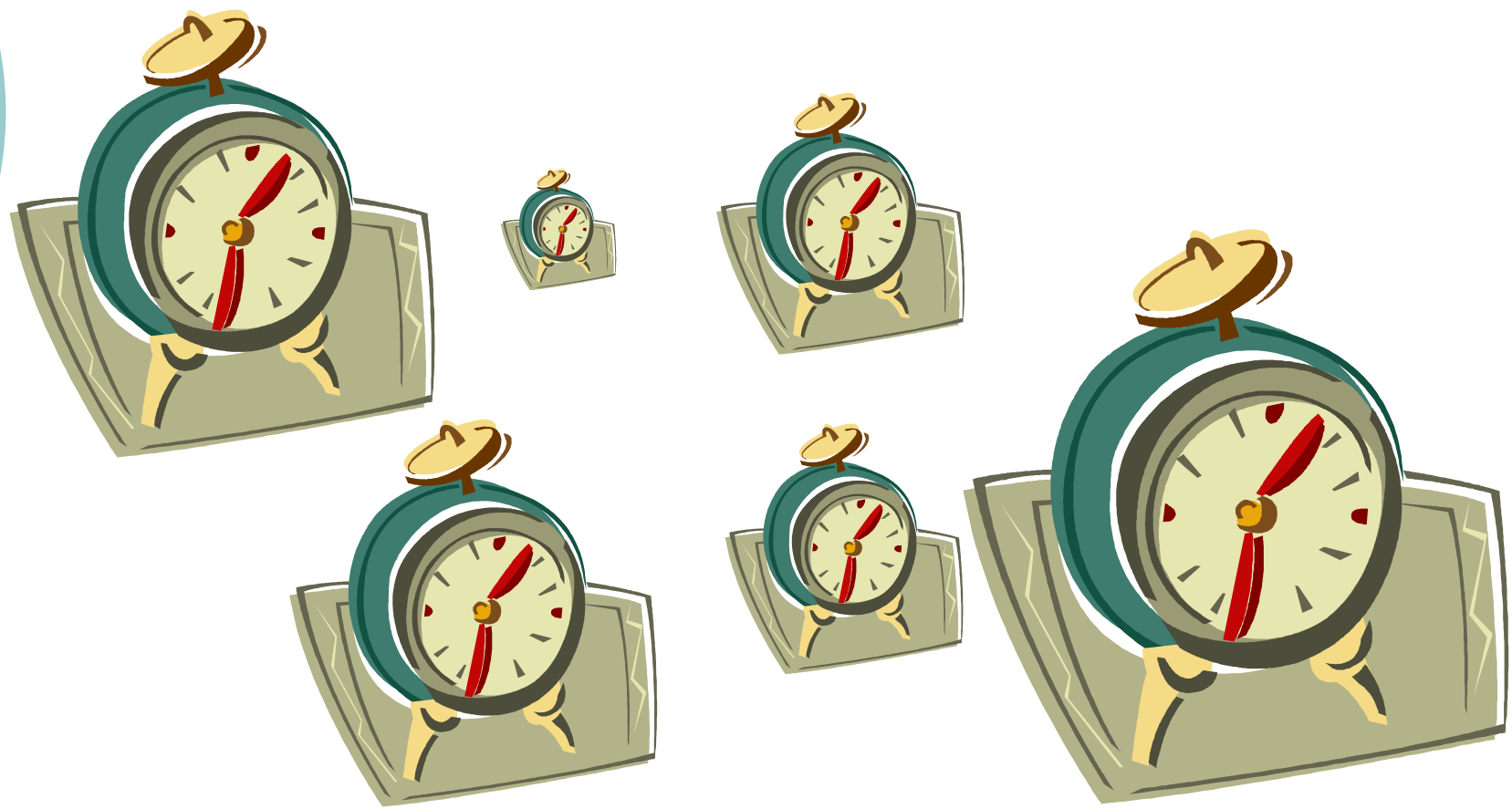
$$\begin{cases} (x-x_A)^2+(y-y_A)^2=d_A^2 \\ (x-x_B)^2+(y-y_B)^2=d_B^2 \\ (x-x_C)^2+(y-y_C)^2=d_C^2 \end{cases}$$



Performance



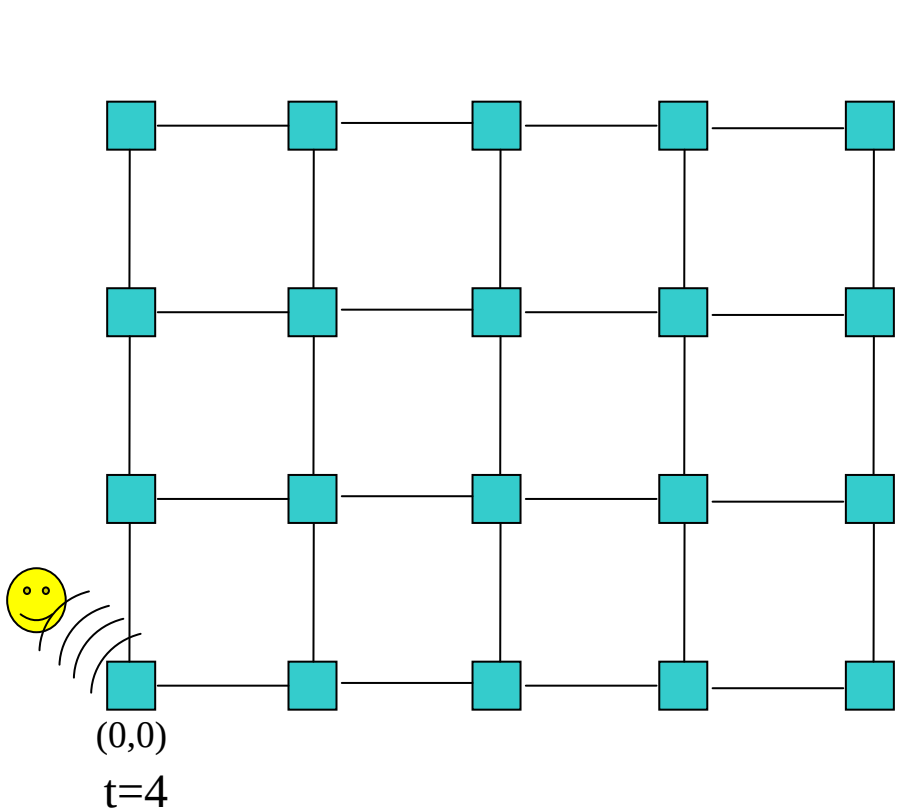
Time Synchronization in Sensor Network

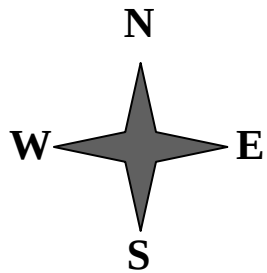


What is Time Synchronization?

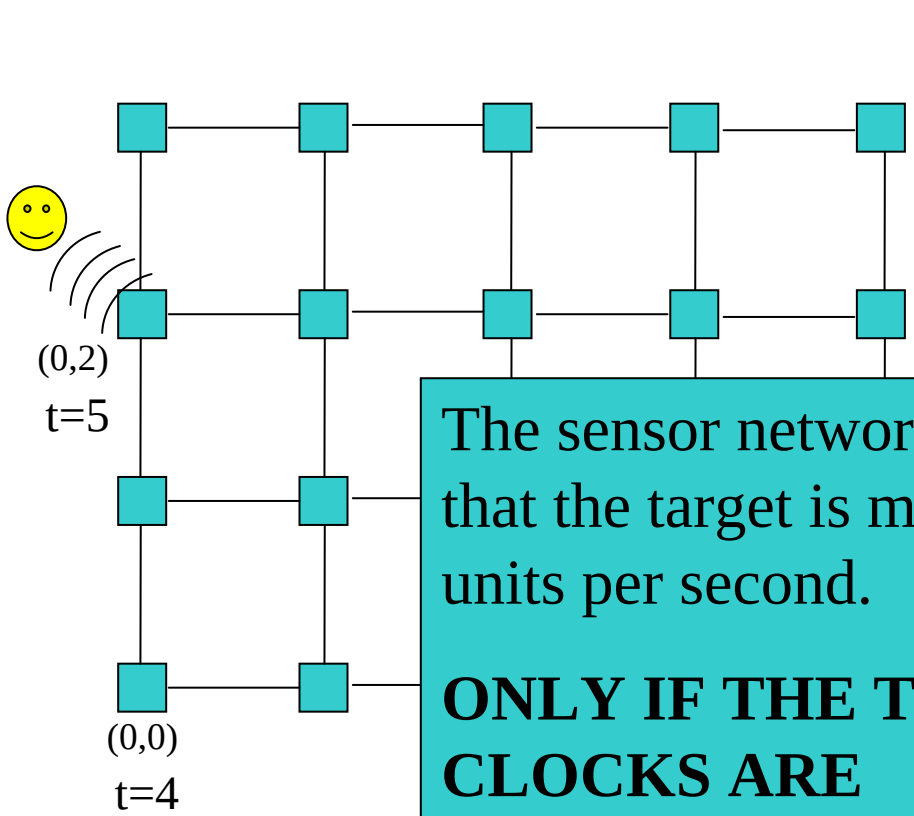
- Time synchronization in a sensor network is the process to have sensor's internal clocks ticking in phase.
- It is critical in sensor network for diverse purposes including sensor data fusion, coordinated actuation, power-efficient duty cycling.

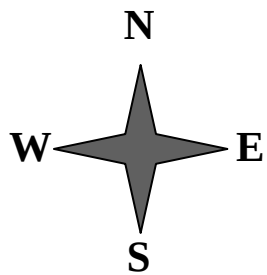
An Example





An Example





The sensor network can conclude that the target is moving north at 2 units per second.

ONLY IF THE TWO SENSORS CLOCKS ARE SYNCHRONIZED!!!!

Overview

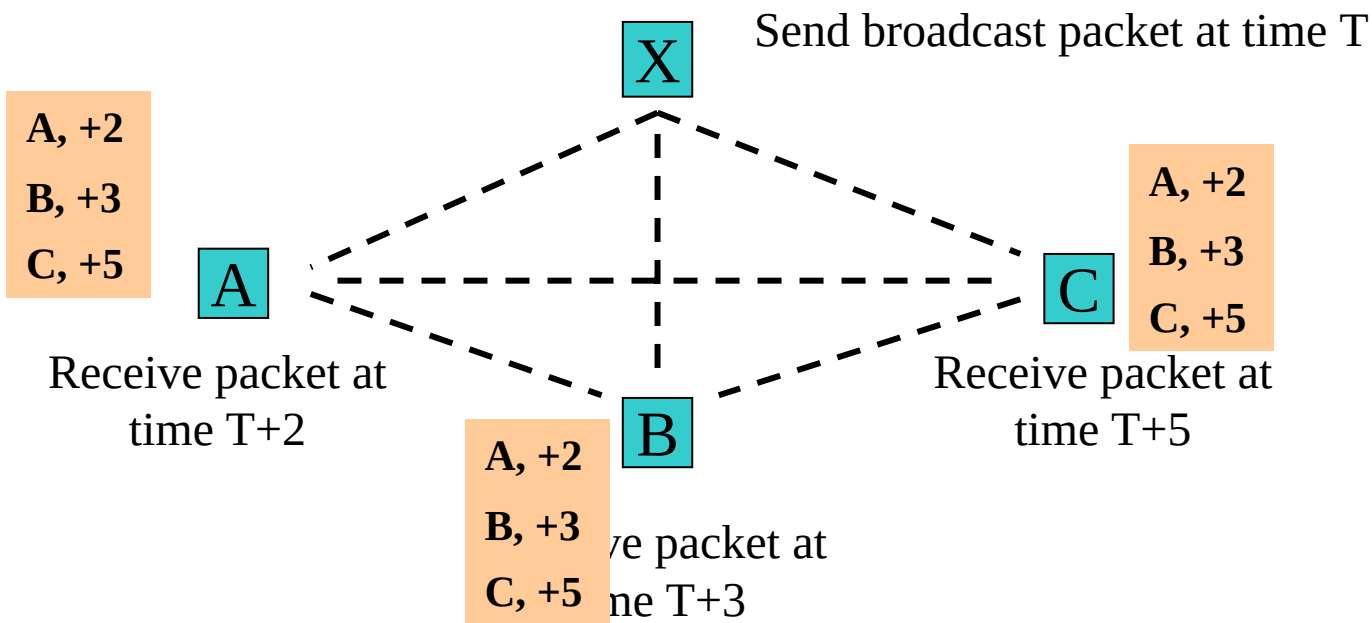
- Reference Broadcast Synchronization (RBS)
- NPT (Network Time Protocol) in amorphous networks

Reference Broadcast Synchronization (RBS)

- Synchronization mechanism developed at UCLA
  - One-hop synchronization
  - Clock skew
  - Multi-hop synchronization

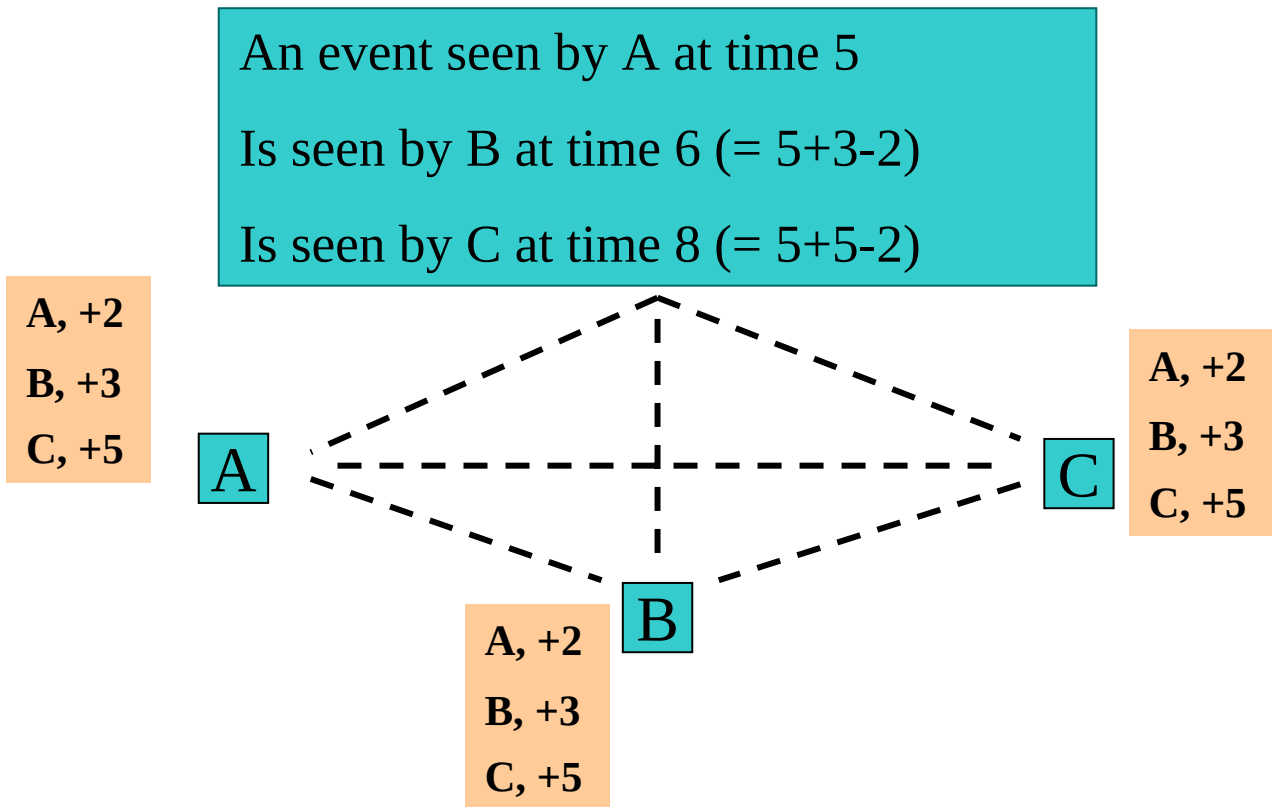
RBS – One Hop Synch

- RBS is based on a broadcast message used to synchronize a set of receivers with one another.



RBS – One Hop Synch

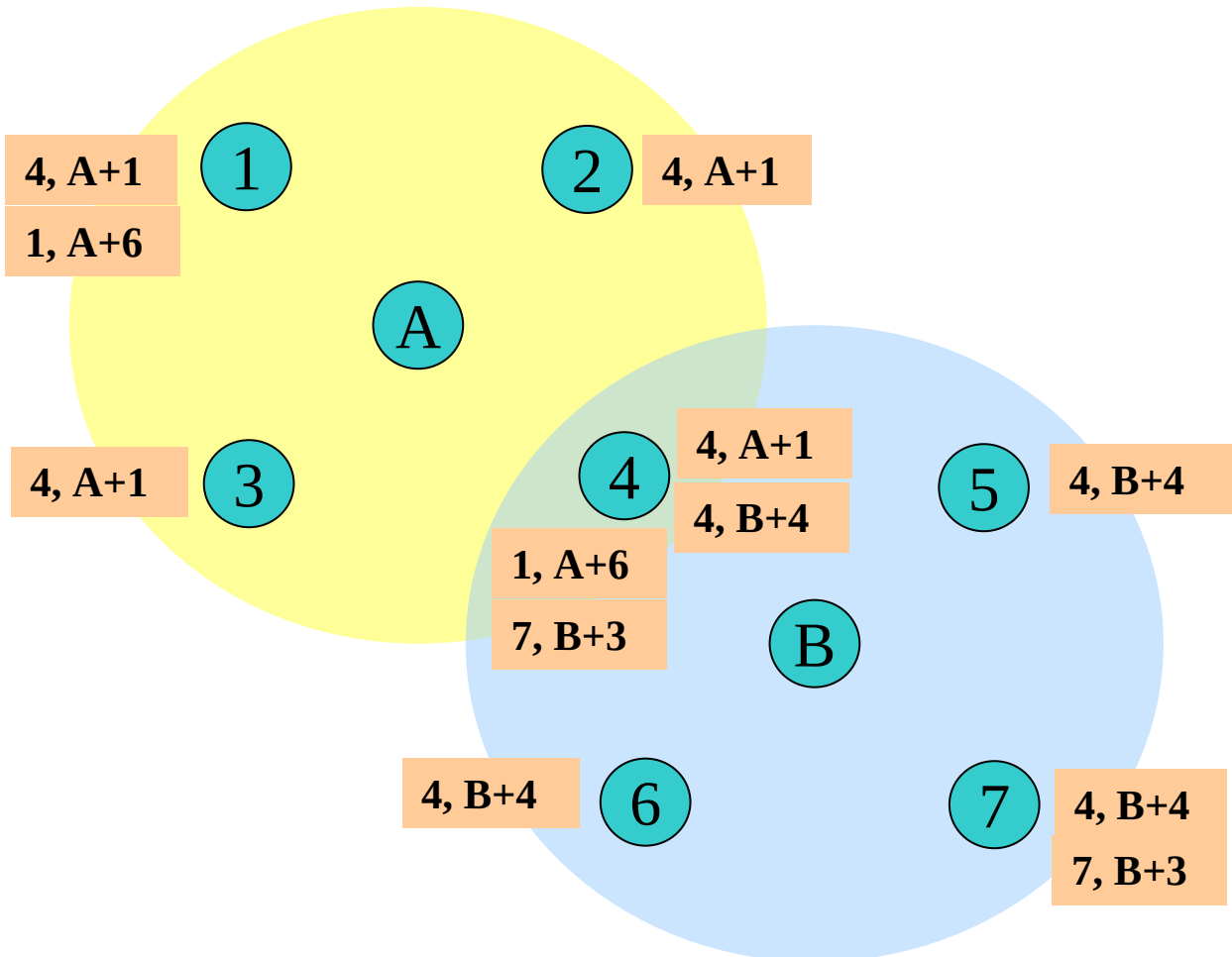
- RBS is based on a broadcast message used to synchronize a set of receivers with one another.



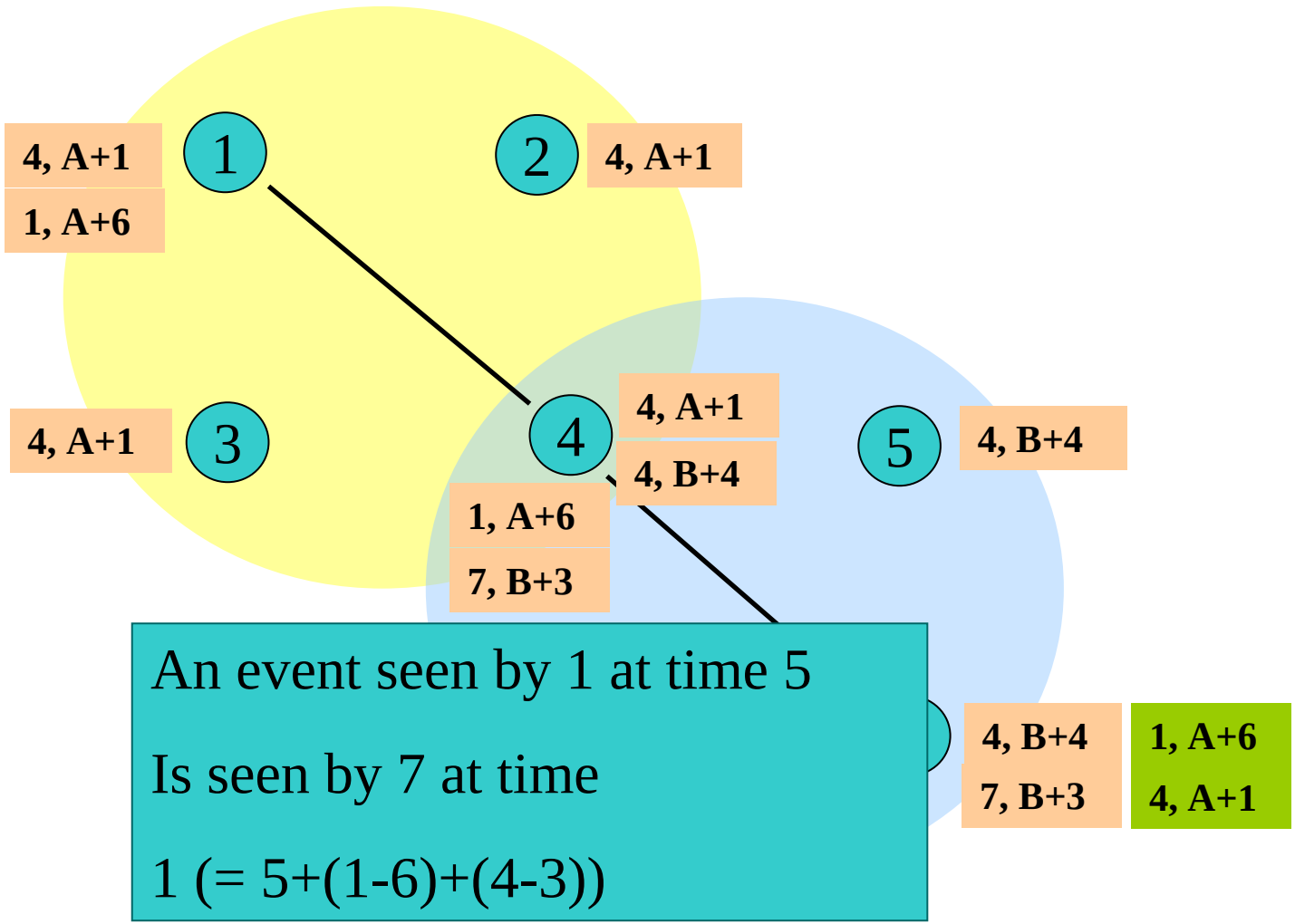
Clock Skew

- Clock skew (sensors' clocks ticking at different rates) rend synchronization soon obsolete…
- Solution
  - Send multiple broadcast synchronization packets
  - Interpolate the packets receive offset time
  - Send the line equation to neighbors, instead of offset
  - Time conversion rates are evaluated on line equations.

RBS – Multi Hop Synch



RBS – Multi Hop Synch



RBS Requirements

- A-Priori synchronization
  - Traffic and load of table-driven routing protocols.
- Post-facto synchronization
  - Same traffic and load on-demand routing protocols

UCLA RBS is based on post-facto synchronization

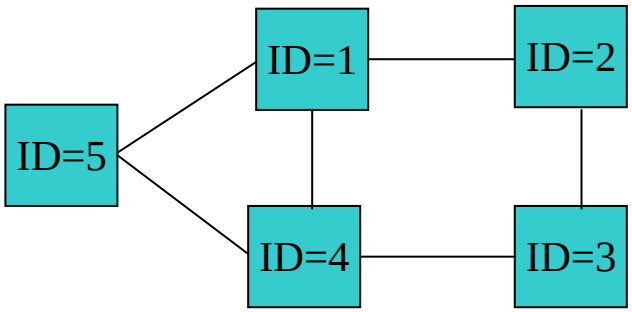
NPT in Amorphous Network

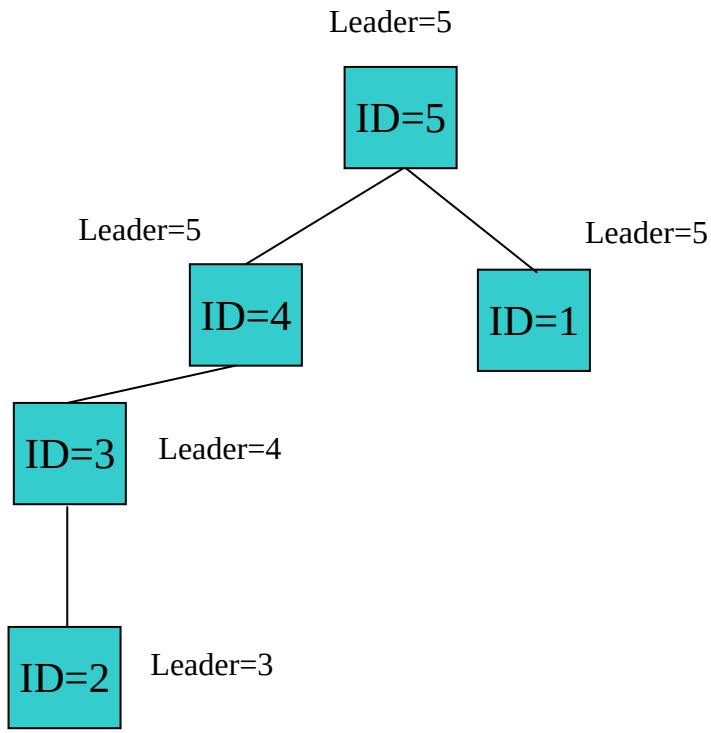
- Proposal from MIT, to overcome some shortcomings of RBS
  - Who sends the broadcast messages to synchronize ?
  - How to achieve a-priori synchronization avoiding network flooding?

Leader Election 1

- Each processor assigns itself an integer ID randomly (large number to avoid collisions), and assumes the leader role.
- Each processor transmits its ID when it spikes.
- If a processor hears an ID higher than its own, it changes its leader to that ID to match and spikes.

Leader Election 2





Tree-based NTP

- The leader constantly broadcasts a synch message
- 1st level nodes synchronize to the leader, and re-propagate the synch message to children with their time-offset information
- 2nd level nodes synchronize with the root, taking into account message received from the parent
- ...

Global Synchronization

- A tree is created with all the nodes synchronized to the root-time.
- The tree is made robust to network changes, by each node constantly checking the neighborhood….
- If the root has some problems
  - The root synchronize to 1st level nodes
  - New leader election…

Performance

- RBS
  - Tested on Compaq IPAQ, IEEE 802.11
  - Mean Error: 6.29µsec to 8.44µsec depending on traffic condition
- NPT in amorphous network
  - Only Simulated
  - Theorems prove that errors increase with the square root of the size of the network.

Global Programming

- Programming a sensor network is very complicated due to the extreme distribution of the algorithms
- Idea:
  - *"…**When programming**, the ability to think and describe goals in terms of high-level abstractions, make possible a complexity that is almost inconceivable to generate by manipulating 1s and 0s. Yet the final computation does happen as bits, and **the compiler** translates from a language that is natural for expressing how to do something, to a low-level execution model that a computer can interpret.*
  - *In a lot of sensor network settings the goal is similar: we would like to be able to **translate complex global goals into local behavior**. However, this translation should not be hand crafted for each goal, but there should be some automatic and engineered mechanism to perform the translation: in other words, something like a **global-to-local compiler**…"*

Global Programming

- Definition of macro-level spatial primitives to address and target sensor nodes.
  - All the nodes on the hill start recording…
- High-level primitives enable to focus on high-level behavior, forgetting low-level detail (communication, synchronization), etc.
- Examples:
  - Spatial Programming (Rutgers University)
  - Macroprogramming Myriads of Sensors (Harvard)

- Such global languages rely on low-level applications and services like the ones presented.

Further Applications

- Working on the above basic services, a number of applications have been built by global programming or direct coding
  - SQL for sensor network
  - Tracking
  - Location-based services
  - Data aggregation and Data fusion
  - …